



resume

Paolo Agliati, 28

“ I am deeply curious about the natural aspects of computation and intelligence. My research concerns the development of bio-plausible (spiking) neural network models”

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Arnhem, Netherlands



Studies

- Universiteit van Amsterdam (2021 - 2023)
MSc - Research Master Brain and Cognitive Sciences
- Milano-Bicocca University (2018 - 2021)
BSc - Biotechnologies

Work experience



2023 - Present

PhD Candidate - Donders Center for Cognition

Radboud University, Nijmegen

Artificial Cognitive Systems Group - Investigating control and dynamic behavior in SNNs



2023

Research Intern - Radboud Universiteit

Radboud University, Nijmegen

Collaboration with the Max Planck Institute to explore the biophysical bases of working memory using spiking neural networks (Python, PyTorch, NEST)

2025

Community Facilitator - Dance instructor

Bora Forró, Amsterdam

2022

Social growth advisor - MUSR

Online

2023

Research Internship - "IMCN"

UvA, Amsterdam

2021

Research Internship - "Rita Levi Montalcini"

Milano-Bicocca Univesity, Milan

Current Projects

INI - Zurich

Collaboration with the Institute of Neuroinformatics (ETH-UZH).
Use of control in SNNs for robotic arm movements

SNNs - Learning to control

Integrating bio-plausible learning rules in the existing SNN control paradigm

University of Bern

Running collaboration on control stability proofs and bio-plausibility of spike coding networks

Dynamics of epilepsy

Interal supervision of a project using the SNN control paradigm to investigate the dynamics of Epilepsy onset

Academic experience

Published paper, PLOS computational biology

Spiking neurons as predictive controllers of linear systems

Invited Talk - SNUFA 2025

Spiking neurons as predictive controllers of linear systems

Poster Presentation

5th International Convention on the Mathematics of Neuroscience and AI, Rome

Oxford machine learning summer school

Organizers: AI for Global Goals, the University of Oxford's Deep Medicine program & CIFAR

UvA Project - Neural Dynamics and Deep Learning

Investigations in working memory: large-scale model of the macaque neocortex

Literature Thesis

Brain-inspired memory implementation in reinforcement learning

Literature Thesis

Development of a 3D model for sporadic Parkinson's disease G2019S-LRRK2 using midbrain organoids

UvA Summer School

Computation in consciousness and Perception - Predictive Coding for Binocular Rivalry